

① **REQUIREMENTS — canonical enterprise-Q&A RAG workload**

~2,000 users · ~10/40 rps · 9.2K-in/300-out · quality ~70B



② **CONSTRAINTS — hard gates**

memory (residency ≤ host) · latency (TTFT/TPOT) · economics (per-host ceiling)



③ **GENERATE CANDIDATES**

(a) 8×H100 + prefix cache · (b) P/D-disaggregated 2-pool · (c) KV-quantized 7B



④ **TEST AGAINST memory · latency · economics**

(a) 8×H100 + prefix cache

MEM ✓ · latency ✓ · cost ✓ | but prefill ~19.8K ≪ ~92K needed

✗ **capacity (needs ~5 hosts)**

(b) P/D-disaggregated 2-pool

MEM ✓ · latency ✓ · 2 hosts + fabric (~2× capex)
but one prefill host ~19.8K ≪ ~92K (same as (a))

✗ **at a single prefill host — prefill pool must scale to ~5; carried to Ch.13-14**

(c) KV-quantized 7B

MEM ✓ · cheapest (1 H100) | but fails the QUALITY gate

✗ **EXCLUDED**

⑤ **OUTCOME — SURVIVORS**

(a) per-request-latency baseline

(b) P/D — prefill pool scales independently (needs ~5 hosts); carried to Ch.13-14

✗ **REJECTED**

(c) fails quality gate

Excluded from evaluation